

IC-41 PERIODIC AND SIMPLE HARMONIC MOTION

4-23-2023

FOR IN-CLASS AND TAKE-HOME LAB KIT

41-1 OBJECTIVES

To study simple harmonic motion, and to verify the equation for time period for a pendulum and a vertical spring-mass system.

41-2 EQUIPMENT

Stand and clamp

Hooked Metallic sphere

Hooked weights

Spring

String

Vernier caliper

Stopwatch / cellphone

Triple beam balance

Meter stick



EQUIPMENT FOR OPTIONAL ADDITIONAL STUDY

Motion sensor

Pasco Capstone software

Interface 850



41-3 THEORY

Whenever a restoring force is created due to displacement of a body from its mean (equilibrium) position, the restoring force tries to bring the body back to the original (mean) position. If the restoring force is proportional to the displacement from the mean position, the resulting vibratory motion is called simple harmonic motion. A linear restoring force may be expressed mathematically by the equation

$$F = -k \Delta x \quad (1)$$

where F is the force (N), Δx (m) is the displacement from the equilibrium point, and k (N/m) is a proportionality constant. The minus sign expresses the restoring nature of the force by indicating that the direction of F is always opposite to the displacement Δx from equilibrium.

For vibratory motion we have the following definitions:

- Amplitude, A : Maximum displacement from equilibrium point on either side (unit: m).
- Cycle: One complete vibration (no units).
- Time Period, T : The time required for one complete cycle (unit: s).
- Frequency, f : The number of cycles completed per second ($= 1/T$) (unit: cycles per second, s^{-1} or hertz).
- Angular frequency, ω : The angle turned in one second. $\omega = 2\pi f = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$ (unit: rad/s)

Further analysis of the vibrations resulting from a linear restoring force shows that the time period for one complete vibration is given by:

$$T = 2\pi \sqrt{\frac{m}{k}} \quad (2)$$

where T is the time period, m is the mass of the vibrating body, and k is the proportionality constant from Equation 1. If m is expressed in kilograms and k in newtons per meter, T will come out in seconds.

The following equations give the displacement, velocity and acceleration of the mass as a function of time:

Displacement:	$\Delta x = A \cos(\omega t + \phi)$	$\Delta x_{\max} = A$
Velocity:	$V = -\omega A \sin(\omega t + \phi)$	$V_{\max} = \omega A = A \sqrt{\frac{k}{m}}$
Acceleration:	$a = -\omega^2 A \cos(\omega t + \phi)$	$a_{\max} = \omega^2 A = \frac{k}{m} A$

Two physical examples of this type of force are considered in the present experiment, the simple pendulum and the spring-mass system.

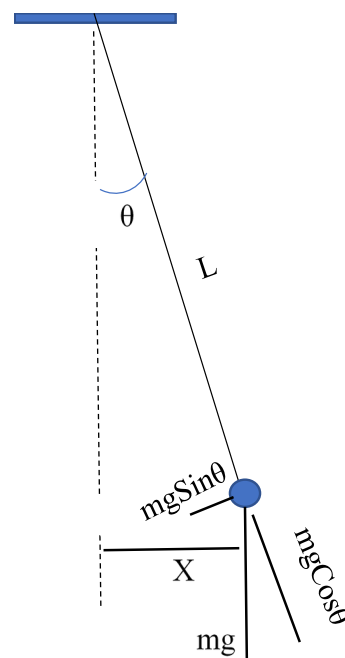
A- SIMPLE PENDULUM

This consists of a concentrated mass suspended at the end of a cord of negligible weight, such as a small metal sphere on a long, thin thread, as shown in Fig. 1. The sphere has been drawn aside to a horizontal distance x , the thread making an angle θ with the vertical. The vertical line through the suspension point is at $X = 0$, which is the equilibrium position. From Fig 1:

For small θ , we have $\sin \theta \approx \theta$, and $m g \sin \theta \approx m g \theta$.

Now $\theta = S/L$, which for small θ , becomes: $\theta = X/L$. The net force then becomes:

$$m g \sin \theta \approx m g \theta = m g X/L = (m g / L) X.$$



In eqn 1, the k becomes the same as mg/L , and the equation for the time period becomes:

$$T = 2\pi \sqrt{\frac{L}{g}} \quad (3)$$

B- SPRING-MASS SYSTEM

This consists of a light spring, to which a mass is attached, such that the mass can move freely. In our case, we will hang the spring vertically, so that the mass can vibrate in a vertical direction.

Ideally, the elongation of a spring is proportional to the stretching force. Therefore, if a mass is hung on a spring, the spring will be stretched by an amount proportional to the weight of the mass. An equilibrium position will be reached at which the spring pulls up on the mass with a force just equal to the mass's weight. If X is the equilibrium position, then when the mass is pulled lower than X by a displacement Δx , there is a restoring force $F = -k \Delta x$, which pulls the mass towards the equilibrium position. This leads to the mass executing simple harmonic motion in the vertical direction.

The time period is given by equation 2:

$$T = 2\pi \sqrt{\frac{m}{k}} \quad (4)$$

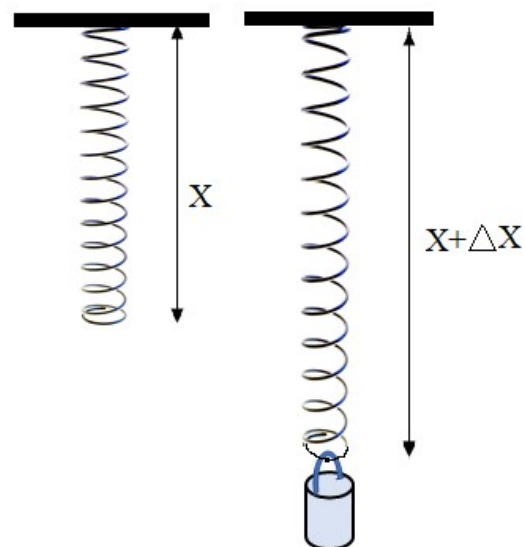


Figure 2: Spring-Mass System

If the mass of the spring is not negligible compared to the attached mass, it must also be included in the computations. It is found that the effective mass m is the attached mass plus one third of the mass of the spring. This is the mass that should be used in the equation for the time period.

41-4 PROCEDURE

A- SIMPLE PENDULUM

- 1- Take a string of length around 2.0 m. Tie a metal sphere to one end, and the other end on a clamp. Measure the distance from the point of support of the string to the center of the metal sphere. This is the length of the pendulum, L . Keep this length to close to about 1.20 m.
- 2- Displace the sphere to one side, so that the string makes an angle of around 5° with the vertical. Release the sphere and let it oscillate. Make sure that the motion is back and forth in a line, and not in an oval path. Start the stopwatch when the sphere is at one extreme end and record the time for 25 vibrations.
- 3- Decrease the length of the pendulum in increments of about 20 to 30 cm each time and find the time for 25 vibrations for each length. Use this data to verify equation 3.

- 4- Use a length of pendulum around 60 cm and find the time for 25 vibrations when the sphere is displaced by about 5° , 30° , and 45° . Use this data to determine the effect of angle (i.e. amplitude) on the time period of the pendulum.
- 5- Use a length of pendulum around 60 cm and find the time for 25 vibrations by using a sphere of a different mass. Use this data to determine the effect of mass of the bob on the time period of the pendulum.

B- SPRING-MASS SYSTEM

- 6- Remove the pendulum, measure mass of the spring, and hang it from the clamp. Measure the position of the lower end of the spring by using a meter stick. This is 'X'. The position can be measured as a height from the floor, or from the table-top, or downwards from the top of the spring.
- 7- Attach different hooked masses and measure the position of the bottom of the spring for each of the attached masses. Use masses in increments of 100 grams to a maximum of about 500 grams (actual values depend on the stiffness and size of the particular spring that you are using. Make sure that the spring does not stretch too much, or it may be damaged). Each new measurement would be 'X+ ΔX '. Use this data to obtain the spring constant k by drawing a graph between ΔX and weight.
- 8- Suspend a mass of 200 grams from the spring and slowly lower it until it stops (this is X). Pull it vertically down by about 5.0 cm and let it go. The spring will undergo simple harmonic motion. Measure the time it takes to complete 25 vibrations. Use this data to verify equation 4.
- 9- Repeat with an initial displacement of 10 cm. The measurements in procedure 7 and 8 are used to study the effect of amplitude on the time period.
- 10- Suspend about 500 grams and measure the time period for an amplitude of 5.0 cm. Use the data from procedure 7 and 9 to study the effect of mass on the time period.

C- OPTIONAL ADDITIONAL STUDY (ONLY FOR IN-CLASS LAB)

11. With the 200 grams mass in place, place a motion sensor under the spring pointing upwards, so that it "sees" the hanging mass.
12. Connect the motion sensor to the Science Workshop. Connect the yellow plug to Digital Channel 1, and the other plug to Digital Channel 2. Connect the Science Workshop to a computer. Click on the PASCO Capstone icon on the computer to open the software for this experiment.
13. Click "Hardware Setup" (on left of screen) to connect to the Science Workshop UI 850 (or the system that you are using as the interface).

14. An image of the Science Workshop will come up. Bring the pointer (mouse) to the Digital Channel 1 on the image and click it. This will open a list of equipment that can be used at that channel. Select Motion Sensor. This will open a window with some graphs on it.
15. Select a graphical output. Select the X-axis as time and set the Y-axis as Position. You can later on change that to Velocity and Acceleration.
16. Pull the mass hanging from the spring about 5 cm, and release, to set up simple harmonic motion.
17. Press the 'Record' button at the lower left of the screen. After collecting data for about 10 vibrations, press the 'stop' button.
18. Click on the "Data Selector" icon to get a colored square on the screen. You can adjust its width and height and move it around the screen. On each of the graphs, move and adjust its size so that a portion of the data that is "good" is inside the box. Then click the icon for Curve Fits and find the best fit for the data. You may have to use "User Defined" equation to get the best fit.
19. The best fit should be a sine curve. Get the value of omega from the curve, and use this to find the Time Period, by: $T = 2\pi / \omega$
20. Change the Y-axis to velocity and acceleration and measure their maximum values. Compare these with the theoretical maximum values.

41-5 CALCULATIONS

- 1- Use the data from procedures 2 and 3, to obtain the measured time period T_m for each length L . Then compare it with the calculated value of T (from equation 3) and find the percent error.
- 2- Use the data of procedure 4 to determine the effect of angle on the time period of the pendulum. Compare this with the theory.
- 3- Use the data of procedure 5 to study the effect of mass on the time period.
- 3- Calculate the mass of the vibrating system = hanging mass plus one third of the mass of the spring.
- 4- Use the data of procedures 6 and 7 to find the elongation of the spring for each of the masses and plot the weight (not mass) on the y-axis versus the elongation on the x-axis. Use linear least square fit to fit a straight line to the data. The slope of this line is the spring constant k .
- 4- Use the data of procedures 8, 9 and 10 to calculate the time period of the spring-mass system. Compare these with the theoretical values and find the percent error.
- 5- For Optional Study, obtain the values of amplitude, maximum velocity and maximum acceleration from the graphs. Compare these with the values calculated from relevant equations in the Theory section.

41-6 PRECAUTIONS:

1. The swing of the pendulum should not be too large. The equation 3 is valid for small displacement.
2. Avoid places where there is an air draft, else it may affect the time period of the pendulum.
3. One complete vibration is when the pendulum goes from one extreme position to the next and comes back to the original position.
4. The pendulum should oscillate in a straight line, not in an elliptical path.
5. In the Spring-Mass system, the amplitude of vibration should not be so large that the spring coils hit each other.
6. While oscillating, the spring should move up and down, not up-down and sideways.
7. The length of the pendulum, angle of swing, masses attached to the spring, number of vibrations are all typical values. The actual values will depend on the space and type of springs available to you.
8. If the time per vibration appears to be large, you can use fewer than 25 vibrations – Use more vibrations if the time period is short, and fewer vibrations if the time period is long.

41-7 PERIODIC AND SIMPLE HARMONIC MOTION

REPORT FORM

A- SIMPLE PENDULUM

Table 1: Effect of length of pendulum:

Length of pendulum L	Number of vibrations N	Measured Time for N vibrations t	Measured Time period $T_m = t / N$	Calculated Time period from Eqn. 3 T_c	Percent error in the two time periods

Average percent error in Time period: _____

Table 2: Effect of amplitude of vibration of pendulum:

Length of pendulum, L : _____

Initial displacement of sphere	Number of vibrations, N	Time for N vibrations, t	Time period, $T = t / N$	Time period calculated from Eqn. 3	Percent error in the two time periods

5°					
30°					
45°					

Table 3: Effect of mass of bob of pendulum:

Mass of bob	Number of vibrations, N	Time for N vibrations, t	Time period, $T = N/t$	Time period calculated from Eqn. 3	Percent error in the two time periods

B- SPRING-MASS SYSTEM

Table 4: Determine Force Constant

Mass suspended from the spring	Force stretching the spring	Position of the bottom of the spring	Elongation, ΔX
0.0 kg			X
0.1 kg			
0.2 kg			
0.3 kg			
0.4 kg			
0.5 kg			

Force constant k of the spring: _____ (attach graph)

Table 5: Verify equation 4

Mass of the spring: _____

Mass suspended from the spring, M	Mass of the vibrating system, m	Amplitude of vibration	Time for 25 vibrations, t	Time Period		Percent error
				Experimental value, $T = t/25$	Calculated value (eqn 4)	

Table 6: Effect of mass and amplitude on the time period

Mass of the spring: _____

Mass suspended from the spring, M	Mass of the vibrating system, m	Amplitude of vibration	Time for 25 vibrations, t	Time Period		Percent error
				Experimental value, $T = t/25$	Calculated value (eqn 4)	
M1=		L1=				
M1=		L2=				
M2=		L1=				
M2=		L2=				

C- DATA FROM ADDITIONAL STUDY

Best fit curve: _____

Maximum Displacement (Amplitude) from software: _____

Angular frequency ω :

from Software: _____

Calculated value: _____

Percent Error: _____

Maximum Velocity

from Software: _____

Calculated value: _____

Percent Error: _____

Maximum Acceleration:

from Software: _____

Calculated value: _____

Percent Error: _____

41-8 REPORT SUBMISSION

Upload the following in the Report for this Lab:

		Points in report
1.	The completely filled up "Report Form". Make sure to include units of measurements.	$2 \times 10 = 20$
2.	Graph for the Spring constant showing the best fit line. Don't forget to label the axes.	$2 \times 5 = 10$
3.	Sample Calculations for the two parts.	$2 \times 5 = 10$
4.	Sources of Error in this experiment. Indicate the major sources of error. Make a List.	5
5.	Discussion of Results	10
	Total	65

"Sources of Error" and "Discussion of Results" are two separate headings

Extra Credits: Optional Study, 10 points

Short video of any one part of the Lab., 5 points

41-9 POINTS TO THINK ABOUT

1. Should the time period of the pendulum depend on the mass of the bob?
2. Why should the amplitude remain small for the pendulum?
3. Why is one third of the mass of the spring added to the hanging mass?
4. What will happen to the time period of the pendulum if the amplitude is too large.
5. What is the Foucault Pendulum?
6. What is the purpose of the pendulum in a grandfather clock?
7. Will the motion of a child on a swing follow the same equations as the pendulum in the lab?
8. What will be the effect of a bob that is not physically small (say its 20 cm in diameter in a one-meter-long pendulum)?
9. Will a 'stiffer' spring have a longer or shorter time period (for the same hanging mass)?
10. What is an advantage of measuring time for a larger number of vibrations?
11. What is a disadvantage of measuring time for a larger number of vibrations?

41-10 SAMPLE DATA**A- SIMPLE PENDULUM**

Length of pendulum, L (m)	Number of vibrations, N	Measured Time for N vibrations, t (s)	Measured Time Period, $T = t / N$ (s)	Calculated Time Period from Eqn. 3 (s)	Percent error in the two time periods
1.0	25	48.3	1.93	2.01	3.8
1.5	25	63.2	2.53	2.46	2.9
2.0	20	55.2	2.76	2.84	2.8
2.5	20	62.8	3.14	3.17	1.1
3.0	10	36.4	3.64	3.48	4.7
3.5	10	38.1	3.81	3.75	1.5

B- SPRING-MASS SYSTEM

(From Cyrus and Redmond CRN 11937)

Mass suspended from the spring	Force stretching the spring	Position of the bottom of the spring	Elongation, ΔX
kg	N	m	m
0.0	0.256	0.605	0.00
0.1	1.727	0.587	0.018
0.2	2.218	0.567	0.038
0.3	3.199	0.528	0.077
0.4	5.161	0.448	0.157
0.5	10.07	0.250	0.355

Force constant k of the spring: 24.7 N/m

(attach graph)

Mass of the spring: 0.0782 kg

Mass suspended from the spring, M	Mass of the vibrating system, m	Amplitude of vibration	Time for 25 vibrations, t	Time Period		Percent error
				Experimental value, $T = t/25$	Calculated value (eqn 4)	
kg	kg	m	s	s	s	
0.4	0.426	0.076	20.60	0.824	0.825	0.061
0.5	0.526	0.107	23.16	0.926	0.916	1.107
1	1.026	0.100	31.71	1.268	1.280	0.878

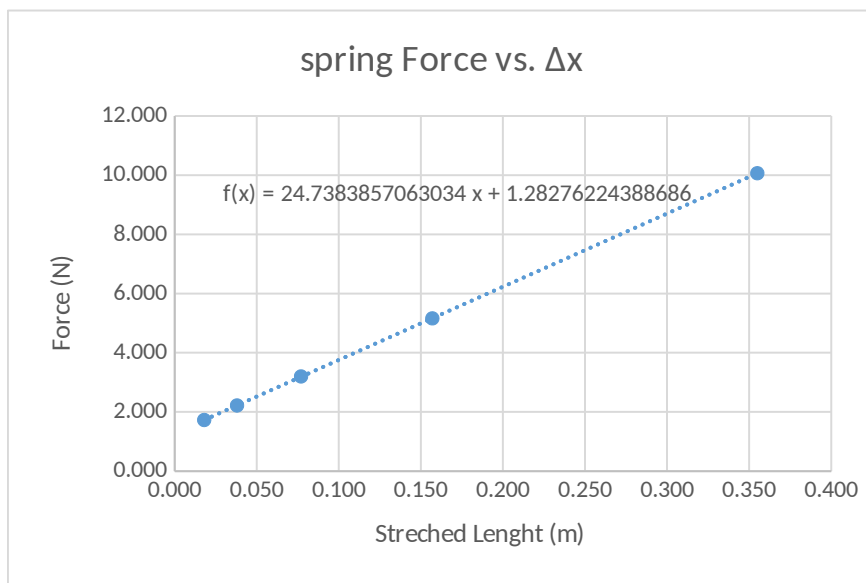


Figure 1 Graph of Force vs. Displacement with slope representing the spring k constant

C- DATA FROM ADDITIONAL STUDY

Best fit curve: $A \sin(\omega t + \phi) + C$

Maximum Displacement (Amplitude) from software: 0.135 [m]

Angular frequency ω :

from Software: 6.74 [rad/s]

Calculated value: 7.029[rad/s] Percent Error: 4%

Maximum Velocity

from Software: 0.859 [m/s]

Calculated value: 0.95 [m/s] Percent Error: 9%

Maximum Acceleration:

from Software: 5.48 [m/s²]

Calculated value: 6.67[m/s²]

Percent Error: 18%

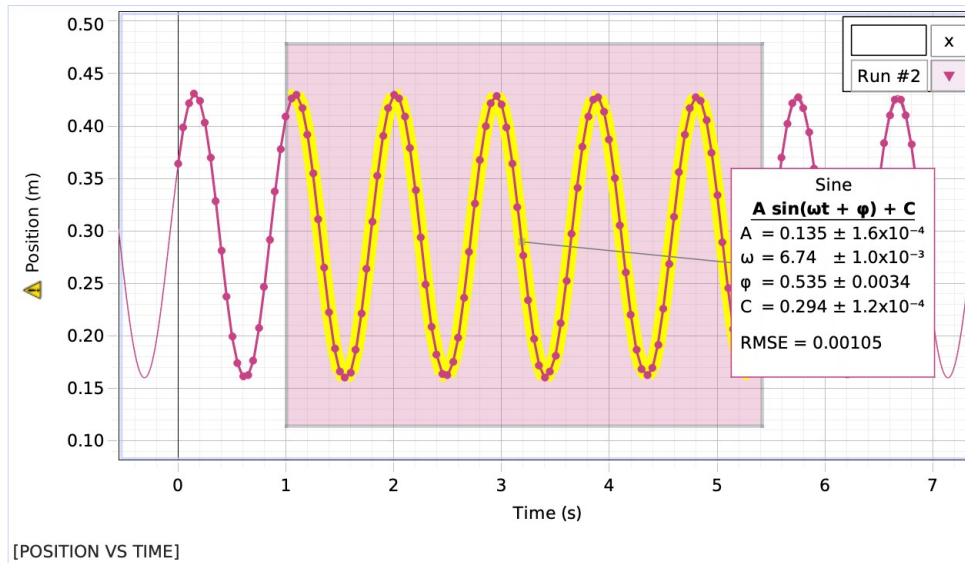


Figure 2 Graph of Recorded data from oscillating spring Position Vs. Time

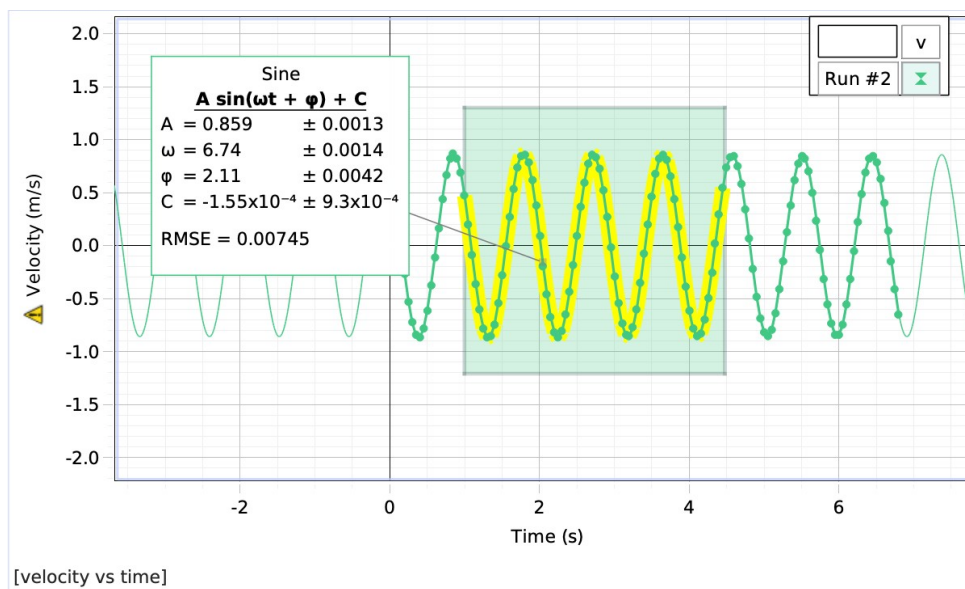


Figure 3 Graph of Recorded data from oscillating spring Velocity Vs. Time

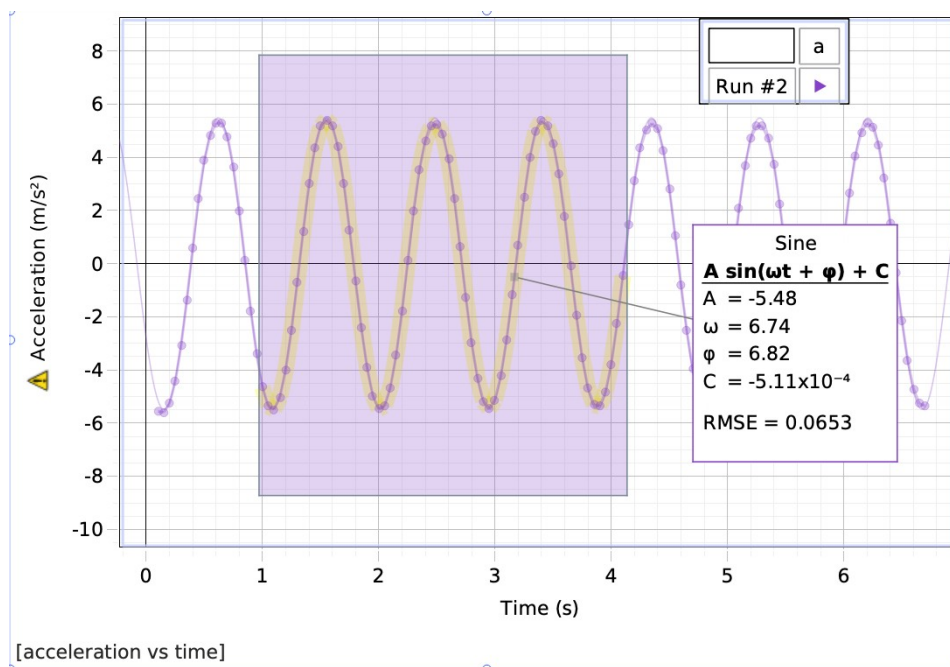


Figure 4 Graph of Recorded data from oscillating spring Acceleration Vs. Time

Sample Calculations

$$(1) \text{ Percent Error} = \frac{\left(6.67 \frac{m}{s^2}\right) - \left(5.48 \frac{m}{s^2}\right)}{6.67 \frac{m}{s^2}} 100 = 18\%$$

$$(2) T_{\text{SpringMass}} = \frac{20.6 s}{25} = 0.824 s$$

$$(3) T = 2\pi \sqrt{\frac{0.426(m)}{24.7\left(\frac{N}{m}\right)}} = 0.825 s$$

$$(4) \Delta x = (A) \cos(\omega t + \phi) \Delta x_{\text{max}} = 0.135(m)$$

$$(5) v = -(A\omega) \sin(\omega t + \phi) V_{\text{max}} = \omega A = 0.135(m) \sqrt{\frac{24.7\left(\frac{N}{m}\right)}{0.5 m}} = 0.95 \left[\frac{m}{s}\right]$$

$$(6) a = -(A\omega^2) \sin(\omega t + \phi) a_{\text{max}} = \left(7.029 \left(\frac{rad}{s}\right)\right)^2 \times (0.135(m)) = 6.67 \left(\frac{m}{s^2}\right)$$